

Anisotropic Energetic Mesh Smoothing with a Metric Tensor Field in Norma

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Abstract

Energetic mesh smoothing (EMS) in Norma minimizes a hyperelastic energy of the mesh, with an ideal reference element per tetrahedron set by a scalar size. This note records the extension of that ideal element to a metric tensor, which prescribes three principal sizes and their directions, and the change in the energy that makes the orientation of the elements part of the objective: the energy is evaluated on the deformation gradient measured in the metric, and the stress and tangent are pulled back to the reference configuration. The formulation, its properties, the treatment of a metric that varies in space, the implementation, the regression tests, and two examples are described.

1 Formulation

The isotropic smoothing energy. For each tetrahedron the ideal reference element $\Omega_{\mathbf{X}}$ is a regular tetrahedron of edge length h , the current element $\Omega_{\mathbf{x}}$ is the element of the mesh, and $\mathbf{F} = \partial \mathbf{x} / \partial \mathbf{X}$ is the deformation gradient of the map between them. The smoothing energy of the mesh is $\Phi = \sum_e V_{\mathbf{X}}^e W(\mathbf{F}_e)$, with W an isotropic hyperelastic energy density (the Seth–Hill family in Norma) and $V_{\mathbf{X}}^e$ the volume of the ideal element. The nodal positions are the unknowns and the ideal elements are fixed during a solve. Because W is isotropic, it depends on $\mathbf{F}$ only through the invariants of $\mathbf{C} = \mathbf{F}^T \mathbf{F}$, and $W(\mathbf{Q} \mathbf{C} \mathbf{Q}^T) = W(\mathbf{C})$ for every rotation $\mathbf{Q}$: rotating the ideal element changes nothing. This is harmless for a regular ideal element, which has no orientation, and fatal for an anisotropic one: an elongated element has zero energy in every orientation, so the energy cannot ask the elongation to point in a prescribed direction.

The metric tensor. An anisotropic target is a symmetric positive definite metric tensor

$$\mathbf{M} = \mathbf{R} \operatorname{diag}\left(\frac{1}{h_1^2}, \frac{1}{h_2^2}, \frac{1}{h_3^2}\right) \mathbf{R}^T, \quad (1)$$

with principal sizes h_i and a rotation $\mathbf{R} = \exp(\hat{\mathbf{v}})$, given by a rotation vector $\mathbf{v}$, whose columns are the principal directions. The length of a vector $\mathbf{e}$ measured in the metric is $\sqrt{\mathbf{e} \cdot \mathbf{M} \mathbf{e}}$, so an edge of length h_i along the i -th principal direction has unit length in the metric. Any factor $\mathbf{F}_M$ with $\mathbf{F}_M^T \mathbf{F}_M = \mathbf{M}$ maps physical vectors to metric-space vectors; Norma uses

$$\mathbf{F}_M = \operatorname{diag}(1/h_1, 1/h_2, 1/h_3) \mathbf{R}^T, \quad \mathbf{F}_M^{-1} = \mathbf{R} \operatorname{diag}(h_1, h_2, h_3). \quad (2)$$

The ideal element is the unit regular tetrahedron $\Omega_{\mathbf{Y}}$ mapped by $\mathbf{F}_M^{-1}$, that is, scaled by h_i along the global axes and then rotated by $\mathbf{R}$: $\mathbf{X} = \mathbf{F}_M^{-1} \mathbf{Y}$. Its volume is $h_1 h_2 h_3 V_{\text{unit}}$ with $V_{\text{unit}} = 1/(6\sqrt{2})$.

The energy in the metric space. The current element is also mapped into the metric space, $\mathbf{y} = \mathbf{F}_M \mathbf{x}$, and the energy is evaluated on the deformation gradient between the unit ideal element and the metric image of the current element,

$$\mathbf{F}_U = \frac{\partial \mathbf{y}}{\partial \mathbf{Y}} = \mathbf{F}_M \mathbf{F} \mathbf{F}_M^{-1}, \quad \Phi = \sum_e V_{\mathbf{X}}^e W(\mathbf{F}_{U,e}). \quad (3)$$

An element has zero energy if and only if its metric image is a unit regular tetrahedron, that is, if it is a unit element for $\mathbf{M}$ in the sense of metric-based meshing. If the current element is the ideal element rotated by $\mathbf{Q}$, then $\mathbf{F} = \mathbf{Q}$ and $\mathbf{C}_U = \mathbf{F}_M^{-T} \mathbf{Q}^T \mathbf{M} \mathbf{Q} \mathbf{F}_M^{-1}$, which equals the identity only when $\mathbf{Q}^T \mathbf{M} \mathbf{Q} = \mathbf{M}$: the orientation is penalized, except for rotations that leave the metric unchanged, as it should be. With equal sizes $\mathbf{F}_M = \mathbf{I}/h$, $\mathbf{F}_U = \mathbf{F}$, and the isotropic energy is recovered exactly.

Stress and tangent. With the ideal volume as the weight, $\Phi_e = V_{\mathbf{X}}^e W(\mathbf{F}_M \mathbf{F} \mathbf{F}_M^{-1})$, and the chain rule gives the first Piola–Kirchhoff stress and the tangent in the reference configuration from those of the isotropic model evaluated at $\mathbf{F}_U$,

$$\mathbf{P} = \mathbf{F}_M^T \mathbf{P}_U \mathbf{F}_M^{-T}, \quad A_{iJkL} = (F_M)_{pi} (F_M^{-1})_{Jq} A_{pqrs}^U (F_M)_{rk} (F_M^{-1})_{Ls}, \quad (4)$$

where $\mathbf{P}_U = \partial W / \partial \mathbf{F}_U$ and $\mathbf{A}^U = \partial \mathbf{P}_U / \partial \mathbf{F}_U$. Nothing else in the assembly changes. The same result follows from the nodal chain rule: with $\mathbf{y}_a = \mathbf{F}_M \mathbf{x}_a$, the nodal forces are $\mathbf{f}_x = \mathbf{F}_M^T \mathbf{f}_y$ and the nodal stiffness blocks $\mathbf{K}_x = \mathbf{F}_M^T \mathbf{K}_y \mathbf{F}_M$.

Properties.

1. The energy does not depend on the factor chosen for $\mathbf{M}$: any other factor is $\mathbf{Q} \mathbf{F}_M$ with $\mathbf{Q}$ a rotation, which enters $\mathbf{F}_U$ as a superimposed rigid rotation and leaves the isotropic W unchanged. The eigenvector signs and the in-plane phase of $\mathbf{R}$ for repeated sizes are therefore irrelevant to the energy.
2. The orientation of the unit tetrahedron $\Omega_{\mathbf{Y}}$ is irrelevant for the same reason: every rotation of the unit tetrahedron mapped by $\mathbf{F}_M^{-1}$ is a zero-energy element. The smoother does not prefer one packing of anisotropic elements over another; the mesh topology decides.
3. $\det \mathbf{F}_U = \det \mathbf{F}$, so the volumetric part of the energy is unchanged by the metric transformation: it compares the element volume with $h_1 h_2 h_3 V_{\text{unit}}$.

2 A metric that varies in space

The metric is evaluated once per element at the centroid of the element in the original mesh, as the isotropic size field is, and held fixed during a solve. The energy is then an exact function of the nodal positions and the assembled force is its exact gradient, which the line search and the convergence tests of the solver require. If the metric followed the current centroid, $\mathbf{F}_M$ would depend on the unknowns and the exact gradient would contain the derivative of $\mathbf{F}_M$, which involves the derivative of the exponential map of the rotation vector; omitting it would make the force inconsistent with the energy. The metric is therefore Lagrangian: an element keeps the target it was assigned where it started. For smoothing, where nodes move a fraction of an element, this is a second-order effect. For large motions, or when smoothing alternates with topological operations, the targets are

refreshed at phase boundaries from the field at the current positions, which is the re-baselining of the adaptivity note ([docs/notes/ems-adaptivity](#)); each phase then minimizes a fixed functional.

The metric takes the pseudo-time t as an argument, so a target can be continued from the sizes of the original mesh to the desired ones over the pseudo-time steps of the quasi-static solve. Because the sizes and the rotation vector are given separately, a linear path in $\log h_i$ and in $\mathbf{v}$ interpolates the sizes geometrically and stays a rotation, which is the interpolation the metric literature recommends.

The volume floor. Smoothing cannot create elements. The isotropic `size field` rule therefore takes the larger of the size field and the edge length of the regular tetrahedron with the volume of the original element, so that no element is asked to shrink below what the mesh topology can accommodate; without it the size field produced slivers. The rule `metric field` applies the same floor to the metric: when $h_1 h_2 h_3 V_{\text{unit}}$ is smaller than the volume of the original element, the three sizes are scaled by the same factor until the volumes match, so the shape and the orientation of the target are preserved. The rule `metric field unrestricted` omits the floor.

3 Implementation

- `MetricField (src/model_types.jl)`: the compiled size and rotation-vector expressions in t, x, y, z and the `restricted` flag; stored on the model as `metric_field`.
- `create_metric_field (src/model.jl)`: reads the `metric field` block under `model (sizes, three expressions; rotation vector, three expressions, optional)` with the same Symbolics pipeline as the size field.
- `create_metric_reference`: evaluates the field at the reference centroid, checks that the sizes are finite and positive, applies the floor, builds $\mathbf{R}$ with `rt_of_rv (src/minitensor.jl)` and returns the ideal element $\mathbf{F}_M^{-1} \mathbf{Y}$ together with $\mathbf{F}_M$ and $\mathbf{F}_M^{-1}$.
- `evaluate_element!`: with a metric field, the ideal element is built by the function above, the constitutive model is evaluated at $\mathbf{F}_U$, and the stress and the tangent are pulled back with (4) by `pull_back_tangent`. Without a metric field the code path is unchanged.
- Input validation knows the new keys (`metric field, sizes, rotation vector`); the reference documentation is in [docs/src/reference/model.md](#). The sampled sizes and rotation vector are written as the nodal variables `size_1..3` and `rotation_1..3`.

4 Tests

`test/smoothing-metric.jl` (test 126) checks:

1. the ideal element and the factor: $\mathbf{F}_M \mathbf{F}_M^{-1} = \mathbf{I}$, $\mathbf{F}_M^T \mathbf{F}_M = \mathbf{M}$ with and without rotation, unit edges of $\mathbf{F}_M \mathbf{X}$, the ideal volume, agreement with the isotropic size field for equal sizes, the floor (volume of the original element recovered, size ratios preserved), evaluation at the reference centroid and at the supplied time, and the input errors;
2. the orientation sensitivity on one element with sizes (0.25, 1, 1) rotated by 0.4 rad about z : the ideal element has zero energy, the same shape rotated by a quarter turn about z has energy above one while the isotropic energy on $\mathbf{F}$ is zero for both, a rotation about the axis

of transverse isotropy of the metric keeps the energy at zero, and $\mathbf{Q}\mathbf{F}_M$ gives the same energy as $\mathbf{F}_M$;

3. the assembled force and stiffness on the cube mesh (`examples/ems/cube`, 7677 elements) under a metric with all six expressions varying in space: central finite differences of the energy match the force to 10^{-5} relative and a finite-difference directional derivative of the force matches the stiffness to 10^{-5} relative;
4. a smoothing run on the cube with a constant metric at 45° in the x - y plane (sizes $0.5H$, $1.5H$, $1.5H$ with H the mean edge length): the energy decreases, the mean deviation of the edge lengths measured in the metric from unity decreases, and no element inverts.

The existing smoothing tests (51, 119) and the input-validation sweep of the examples (122) pass unchanged.

5 Examples

Cube, constant metric at 45° (`examples/ems/cube/cube-metric.yaml`). Sizes 0.05, 0.1414, 0.1414 (the volume of the mesh size 0.1), first principal direction at 45° in the x - y plane, unrestricted rule, nodes fixed in the normal direction on the faces, L-BFGS. Over the edges of the mesh the mean of $|\|\mathbf{F}_M\mathbf{e}\| - 1|$ goes from 0.44 to 0.31, the mean edge component along the small direction from 0.055 to 0.042 (target 0.05), and the mean component across it from 0.085 to 0.098. The number of elements is fixed, so the metric cannot be met everywhere; the elements rotate and stretch toward it.

Tube, graded axial size (`examples/ems/tube/tube-metric.yaml`). The rotation vector $(0, -\pi/2, 0)$ carries the first principal direction onto the axis of the tube; the axial size grows linearly from 0.06 at the bottom to 0.18 at the top while the radial and circumferential sizes stay at the mesh size 0.12, restricted rule, nodes sliding on the four analytic surfaces. The mean axial size equals the mesh size, so the grading can be realized by sliding without a change of topology. The mean of $|\|\mathbf{F}_M\mathbf{e}\| - 1|$ over the edges goes from 0.18 to 0.12, and the mean length of the axial edges (edges with more than 70% of their length along z) in the four quarters of the height, from bottom to top, goes from 0.105, 0.109, 0.109, 0.105 to 0.075, 0.110, 0.127, 0.144, against targets of 0.075, 0.105, 0.135, 0.165 at the quarter centers. The largest nodal motion is 0.28, about two element sizes, with no element inversion.

What smoothing cannot do. A target that changes the number of elements across the wall or around the circumference, such as a radial size of a quarter of the wall thickness on a mesh with one element through the wall, is not reachable by smoothing: the mesh barely moves and the misfit does not decrease. Such targets require the topological operations of the adaptivity note.

6 Next steps

1. Nodal metrics carried by the mesh (six nodal variables: the logarithms of the sizes and the rotation vector), interpolated to the centroid with constant weights, and re-sampled from the field at phase boundaries.

2. Metrics given as full tensors: recovery of sizes and rotation vector by eigendecomposition with a consistent frame (eigenvector signs by continuity, a fixed in-plane phase for repeated sizes), with the log-Euclidean interpolation of the tensor as the fallback.
3. Coupling with the topological operations of the adaptivity note, so that targets the smoother cannot reach are met by refinement and coarsening in the metric.